

OPTIMIZATION FUNDAMENTALS

Standard Forms: **General:** $\min f(x)$ s.t. $g_i(x) \leq b_i, h_j(x) = c_j, x \in X$ **LP Standard:** $\min c^T x$ s.t. $Ax = b, x \geq 0$ **LP Canonical:** $\min c^T x$ s.t. $Ax \leq b, x \geq 0$

Linear Algebra Essentials: **Rank:** $\text{rank}(A) = \#$ linearly independent rows/cols **Full rank:** $m \times n$ matrix, $\text{rank}(A) = \min(m, n)$ **Matrix inverse:** $AA^{-1} = I$ (exists iff A square & full rank) **Determinant:** $\det(A) \neq 0 \Leftrightarrow A$ invertible **Eigenvalues:** $Av = \lambda v$ where $\det(A - \lambda I) = 0$

System Solutions $Ax = b$: $m = n, \text{rank}(A) = n$: Unique solution $x = A^{-1}b$ $m < n$: Infinite solutions (underdetermined) $m > n, \text{rank}(A) = n$: Least squares $x = (A^T A)^{-1} A^T b$ $\text{rank}(A) < \text{rank}([A|b])$: No solution (inconsistent)

Gradients & Hessians: $\nabla f = [\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}]^T$ $\nabla^2 f = [\frac{\partial^2 f}{\partial x_i \partial x_j}]$ (Hessian matrix) **Chain rule:** $\nabla_x f(g(x)) = \nabla g(x)^T \nabla_y f(g(x))$ **Product rule:** $\nabla(fg) = f\nabla g + g\nabla f$

Taylor Expansions: **1st order:** $f(x) \approx f(x_0) + \nabla f(x_0)^T (x - x_0)$ **2nd order:** $f(x) \approx f(x_0) + \nabla f(x_0)^T (x - x_0) + \frac{1}{2} (x - x_0)^T \nabla^2 f(x_0) (x - x_0)$

Optimality Conditions (Unconstrained): **1st order necessary:** $\nabla f(x^*) = 0$ **2nd order sufficient:** - Min: $\nabla f(x^*) = 0$ & $\nabla^2 f(x^*) \succ 0$ (PD) - Max: $\nabla f(x^*) = 0$ & $\nabla^2 f(x^*) \prec 0$ (ND)

Matrix Definiteness: **PD** ($A \succ 0$): $x^T A x > 0 \forall x \neq 0$, all $\lambda_i > 0$ **PSD** ($A \succeq 0$): $x^T A x \geq 0 \forall x$, all $\lambda_i \geq 0$ **ND** ($A \prec 0$): $x^T A x < 0 \forall x \neq 0$, all $\lambda_i < 0$ **NSD** ($A \preceq 0$): $x^T A x \leq 0 \forall x$, all $\lambda_i \leq 0$ **Indefinite:** Mixed signs in eigenvalues

PSD Tests: 1. All eigenvalues ≥ 0 2. All leading principal minors ≥ 0 3. $x^T A x \geq 0$ for all x 4. $A = B^T B$ for some matrix B

CONVEXITY THEORY

Convex Sets: Definition: S convex if $\lambda x + (1 - \lambda)y \in S \forall x, y \in S, \lambda \in [0, 1]$

Examples: - Hyperplane: $\{x|a^T x = b\}$ - Halfspace: $\{x|a^T x \leq b\}$ - Ball: $\{x|\|x - c\| \leq r\}$ - Ellipsoid: $\{x|(x - c)^T P^{-1}(x - c) \leq 1\}, P \succ 0$ - Polyhedron: $\{x|Ax \leq b\}$ - Simplex: $\{x|\sum x_i = 1, x_i \geq 0\}$

Operations preserving convexity: - Intersection: $S_1 \cap S_2$ convex if S_1, S_2 convex - Affine transformation: $\{Ax + b|x \in S\}$ convex if S convex - Cartesian product: $S_1 \times S_2$ convex if S_1, S_2 convex

Convex Functions: Definition: f convex if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$

Tests for Convexity: 1. **Definition:** Check convex combination property 2. **1st order:** $f(y) \geq f(x) + \nabla f(x)^T (y - x) \forall x, y$ 3. **2nd order:** $\nabla^2 f(x) \succeq 0 \forall x$

Common Convex Functions: - Linear: $a^T x + b$ - Quadratic: $\frac{1}{2} x^T P x + q^T x + r$ where $P \succeq 0$ - Exponential: $e^{ax}, a \in \mathbb{R}$ - Power: x^p for $p \geq 1, x \geq 0$ - Logarithmic: $-\log(x)$ for $x > 0$ - Norms: $\|x\|_p$ for $p \geq 1$ - Max function: $\max\{f_1(x), f_2(x)\}$ if f_i convex

Convexity Preservation: - **Nonnegative combination:** $\sum \alpha_i f_i(x), \alpha_i \geq 0, f_i$ convex - **Pointwise maximum:** $\max_i f_i(x)$ if f_i convex - **Composition:** $g(f(x))$ convex if g convex increasing, f convex - **Perspective:** $tf(x/t)$ convex if f convex, $t > 0$

Concave Functions: f concave $\Leftrightarrow -f$ convex **Examples:** $\log(x), \sqrt{x}, x^p$ for $0 < p < 1$

CONVEX OPTIMIZATION

Problem Classification: **Convex Problem:** $\min f(x)$ s.t. $g_i(x) \leq b_i, h_j(x) = c_j$ where f, g_i convex, h_j linear (affine)

Key Property: Local minimum = Global minimum

Level Sets: $L_\alpha = \{x|f(x) \leq \alpha\}$ (sublevel set) If f convex, then L_α convex $\forall \alpha$

Epigraph: $\text{epi}(f) = \{(x, t)|f(x) \leq t\}$ f convex $\Leftrightarrow \text{epi}(f)$ convex

LINEAR PROGRAMMING

Standard Form Conversion: **Max to Min:** $\max c^T x = -\min(-c^T x)$ **Free variables:** $x = x^+ - x^-, x^+, x^- \geq 0$ **Inequality to equality:** Add slack: $\leq$ becomes $=$ with $s \geq 0$ **constraints:** Subtract surplus: $\geq$ becomes $=$ with $s \geq 0$

Fundamental Theorem: If LP has optimal solution, then $\exists$ optimal BFS Optimal value achieved at extreme point of feasible region

Basic Solutions: Given $Ax = b, A \in \mathbb{R}^{m \times n}, m < n$: **Basic solution:** Choose m linearly independent columns (basis) Set $n - m$ nonbasic variables to 0, solve for basic variables **BFS:** Basic solution that is feasible ($x \geq 0$) **Degenerate BFS:** Some basic variables = 0

Extreme Points & Rays: **Extreme point:** Cannot be written as convex combination of other points **Extreme ray:** Unbounded direction from feasible region **Fundamental Theorem:** BFS $\Leftrightarrow$ Extreme point

Simplex Method: 1. Start with initial BFS 2. Check optimality: $c^T - c_B^T B^{-1} A \leq 0$? (reduced costs) 3. Choose entering variable: Most negative reduced cost 4. Minimum ratio test: $\theta = \min\{x_{B_i}/y_i | y_i > 0\}$ 5. Choose leaving variable: Variable achieving minimum ratio 6. Update basis and repeat

Reduced costs: $\bar{c}_j = c_j - c_B^T B^{-1} A_j$ **Optimality:** $\bar{c} \leq 0$ for minimization **Unboundedness:** Some $\bar{c}_j < 0$ but all $y_j \leq 0$

Degeneracy & Cycling: **Degeneracy:** $\theta = 0$ in ratio test **Bland's rule:** Choose smallest index to avoid cycling

DUALITY THEORY

Primal-Dual Pairs: **Primal:** $\min c^T x$ s.t. $Ax = b, x \geq 0$ **Dual:** $\max b^T y$ s.t. $A^T y \leq c$

General form:	Primal		Dual
	$\min c^T x$		$\max b^T y$
	$Ax \geq b$	$\Leftrightarrow$	$y \geq 0$
	$Ax = b$	$\Leftrightarrow$	y free
	$x \geq 0$	$\Leftrightarrow$	$A^T y \leq c$
	x free	$\Leftrightarrow$	$A^T y = c$

Duality Theorems: **Weak duality:** $b^T y \leq c^T x \forall$ feasible x, y **Strong duality:** If primal has optimal solution, then dual has optimal solution with same objective value **Complementary slackness:** At optimality: - $x_j > 0 \Rightarrow \sum_i y_i A_{ij} = c_j$ - $y_i > 0 \Rightarrow \sum_j A_{ij} x_j = b_i$

TRANSPORTATION & ASSIGNMENT

Transportation Problem: $\min \sum_{i,j} c_{ij}x_{ij}$ s.t. $\sum_j x_{ij} = s_i$ (supply), $\sum_i x_{ij} = d_j$ (demand), $x_{ij} \geq 0$

Balanced: $\sum s_i = \sum d_j$ **Feasibility:** Problem feasible iff $\sum s_i = \sum d_j$ **Basic variables:** $m + n - 1$ in BFS **Degeneracy:** When $< m + n - 1$ positive variables

Assignment Problem: Special case: $s_i = d_j = 1, x_{ij} \in \{0,1\}$ LP relaxation gives integer solution automatically

SENSITIVITY ANALYSIS

RHS Changes: Current solution remains optimal for $b' = b + \Delta b$ if: $B^{-1}(b + \Delta b) \geq 0$

Cost Coefficient Changes: Current basis remains optimal for $c' = c + \Delta c$ if: $\bar{c} + \Delta \bar{c} \leq 0$ where $\Delta \bar{c} = \Delta c - c_B^T B^{-1} A$

Shadow Prices: Dual optimal solution y^* gives shadow prices Change in optimal value per unit increase in RHS

NETWORK FLOWS

Max Flow Problem: $\max f$ s.t. flow conservation at each node Capacity constraints: $x_{ij} \leq u_{ij}$ **Max-flow min-cut theorem:** Max flow = Min cut capacity

Min Cost Flow: $\min \sum c_{ij}x_{ij}$ s.t. flow conservation, capacity constraints Generalizes transportation and assignment problems

Shortest Path: Special case of min cost flow with unit supply/demand Dijkstra's algorithm for non-negative weights Bellman-Ford for general weights

NONLINEAR PROGRAMMING

KKT Conditions: For $\min f(x)$ s.t. $g_i(x) \leq 0, h_j(x) = 0$:

Stationarity: $\nabla f(x^*) + \sum \lambda_i \nabla g_i(x^*) + \sum \mu_j \nabla h_j(x^*) = 0$ **Primal feasibility:** $g_i(x^*) \leq 0, h_j(x^*) = 0$ **Dual feasibility:** $\lambda_i \geq 0$ **Complementary slackness:** $\lambda_i g_i(x^*) = 0$

Constraint Qualifications: **LICQ:** $\nabla g_i(x^*), \nabla h_j(x^*)$ linearly independent **MFCQ:** Mangasarian-Fromovitz CQ **Slater's condition:** For convex problems

Second-Order Conditions: **SOSC:** $\nabla^2 L(x^*, \lambda^*, \mu^*)$ PD on critical cone

LAGRANGIAN DUALITY

Lagrangian Function: $L(x, \lambda, \mu) = f(x) + \sum \lambda_i g_i(x) + \sum \mu_j h_j(x)$

Dual Function: $q(\lambda, \mu) = \inf_x L(x, \lambda, \mu)$ Always concave in (λ, μ)

Dual Problem: $\max q(\lambda, \mu)$ s.t. $\lambda \geq 0$ **Weak duality:** $q(\lambda, \mu) \leq p^*$ always **Strong duality:** $q^* = p^*$ under certain conditions

SPECIAL FUNCTIONS & FORMULATIONS

Absolute Value Linearization: $\min |x|$ becomes: $\min t$ s.t. $t \geq x, t \geq -x$

$\min \sum |x_i|$ becomes: $\min \sum t_i$ s.t. $t_i \geq x_i, t_i \geq -x_i \forall i$

Max/Min Linearization: $\min \max\{f_1(x), f_2(x)\}$ becomes: $\min t$ s.t. $t \geq f_1(x), t \geq f_2(x)$

$\max \min\{f_1(x), f_2(x)\}$ becomes: $\max t$ s.t. $t \leq f_1(x), t \leq f_2(x)$

Piecewise Linear Functions: $f(x) = \max\{a_1x + b_1, a_2x + b_2, \dots\}$: $\min f(x)$ becomes $\min t$ s.t. $t \geq a_i x + b_i \forall i$

SOS1 Variables: At most one variable nonzero: $\sum x_i \leq 1, x_i \geq 0$

EXAM-SPECIFIC CALCULATIONS

Matrix Rank Calculation: Row reduce to echelon form, count nonzero rows For $A =$

$$\begin{bmatrix} 0 & 3 & -2 & -1 \\ 4 & -9 & 1 & 3 \\ -6 & -15 & -1 & 5 \end{bmatrix} : \text{Use Gaussian elimination to find rank} = 2$$

Gradient Examples: $f(x_1, x_2) = \sqrt{x_1 - 1} + (x_2 - 1)^{1/4}$ at $(3, 2)$: $\frac{\partial f}{\partial x_1} = \frac{1}{2\sqrt{x_1 - 1}} = \frac{1}{2\sqrt{2}}$

$$\frac{\partial f}{\partial x_2} = \frac{1}{4}(x_2 - 1)^{-3/4} = \frac{1}{4}$$

Hessian Examples: $f(x_1, x_2) = 2(x_1 - 3)^2 + 6x_1x_2 + 2(x_2 + 3)^2 + 6x_1 + 2x_2$ at $(2, 3)$:

$$H = \begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$$

Convexity Tests: Check if $\nabla^2 f \succeq 0$: For $H = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$: PSD iff $a \geq 0, \det(H) \geq 0$

Set Properties: $X = \{x \in [-1, 1]^5 : \sum x_i \geq -2\}$: - **Bounded:** Yes (box constraints) - **Closed:** Yes (closed inequalities) - **Convex:** Yes (box + linear constraint)

PROBLEM TYPES & OUTCOMES

LP Outcomes: 1. **Optimal:** Finite optimal value 2. **Unbounded:** Objective can be made arbitrarily good 3. **Infeasible:** No point satisfies all constraints

Feasibility Tests: Transportation problem infeasible if: $\sum s_i < \max\{d_j\}$ (total supply $\neq$ max single demand)

Boundedness Tests: Set is bounded if can be contained in some ball of finite radius Polyhedron bounded iff no extreme rays exist

DECISION VARIABLES & MODELING

Here-and-Now vs Wait-and-See: **Here-and-now:** Decision before uncertainty realized

Wait-and-see: Decision after uncertainty realized Wait-and-see always gives better or equal objective value

Variable Types: **Continuous:** $x \in \mathbb{R}$ **Integer:** $x \in \mathbb{Z}$ **Binary:** $x \in \{0,1\}$ **Semi-continuous:** $x = 0$ or $x \in [l, u]$

KEY THEOREMS & RESULTS

Fundamental Theorem of LP: If LP has optimal solution, then has optimal BFS Every bounded polyhedron has finite number of extreme points

Caratheodory's Theorem: Any point in convex hull of S can be expressed as convex combination of at most $d + 1$ points from S (where d is dimension)

Farkas' Lemma: Either $Ax = b, x \geq 0$ has solution, or $A^T y \geq 0, b^T y < 0$ has solution, but not both

COMPUTATIONAL NOTES

Complexity: **LP:** Polynomial time (interior point methods) **Simplex:** Exponential worst case, polynomial average **Integer programming:** NP-hard **Convex optimization:** Polynomial time

Common Mistakes: - Confusing convex function with concave function - Wrong constraint qualification application - Incorrect dual formulation - Matrix definiteness errors - Sign errors in KKT conditions